Econ 372 — Homework 3: Hospital Competition and Pricing

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Setup and data cleaning

```
Raw rows: 56942
year
1997      16
1998    1967
1999    3186
2000    3150
2001    3090
2002    3025
2003    2977
2004    2973
2005    2853
2006    2817
2007    2813
2008    2775
2009    2736
2010    2743
2011    2690
2012    2733
2013    2690
2014    2657
2015    2658
2016    2659
2017    2629
2018    1105

Dropped as data errors:
 provider_number       city state  year  beds  tot_discharges       price
          210005  FREDERICK    MD  2000   260      16013780.0    1.953267
          340141 WILMINGTON    NC  2013   598        155781.0 1305.391045

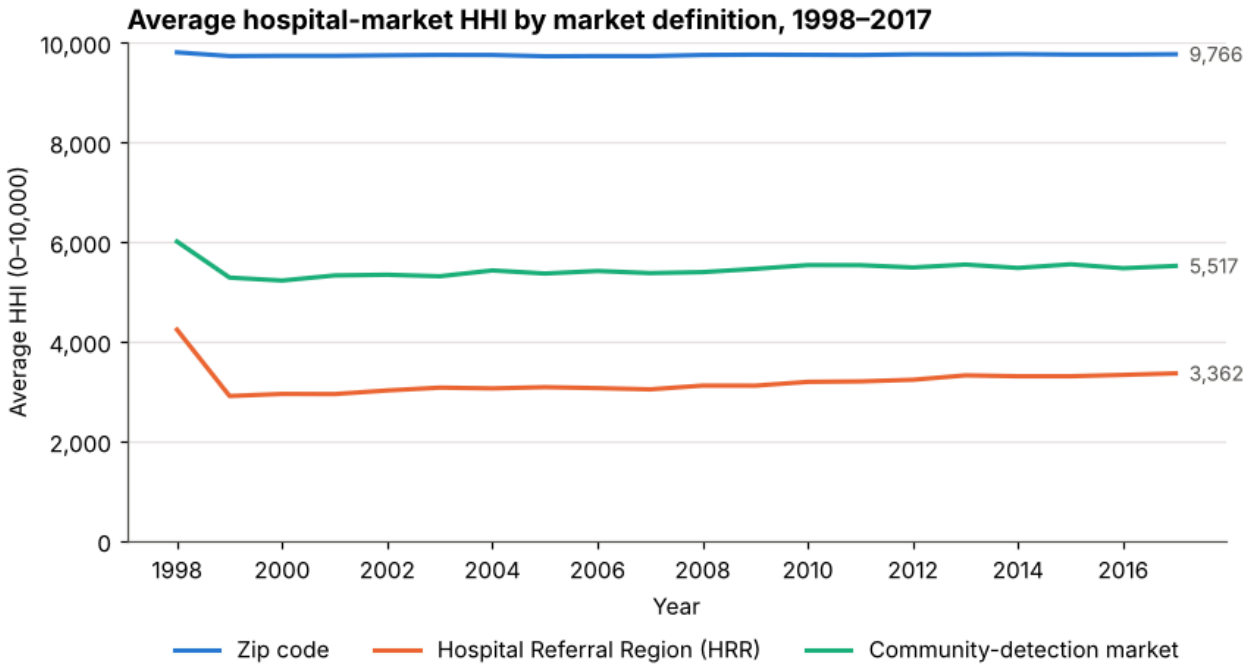
Rows after cleaning: 55819
Rows missing each market identifier:
zip      24
hrr    1580
mkt    6634
```

Hospitals with a missing identifier for a given market definition are left out of that definition only (for example, a hospital with no `mkt` value is still used for the zip and HRR calculations).

Question 1 — Market definition and concentration

1(a) HHI under three market definitions

year	Zip code	Hospital Referral Region (HRR)	Community-detection market
1998	9803	4238	6006
1999	9729	2906	5282
2000	9734	2948	5224
2001	9734	2946	5326
2002	9743	3018	5340
2003	9752	3074	5310
2004	9750	3060	5426
2005	9724	3084	5365
2006	9728	3065	5415
2007	9728	3040	5373
2008	9751	3115	5393
2009	9756	3114	5458
2010	9754	3190	5533
2011	9749	3199	5532
2012	9763	3232	5486
2013	9762	3321	5544
2014	9768	3303	5477
2015	9759	3303	5549
2016	9759	3330	5471
2017	9766	3362	5517



	Markets in 2017	Median hospitals per market	Share of markets with one hospital
Zip code	2478.0	1.0	0.94
Hospital Referral Region (HRR)	303.0	5.0	0.03
Community-detection market	551.0	3.0	0.27

Answer. Measured concentration depends heavily on how the market is drawn. With zip codes as markets, average HHI is about 9,750 in every year (9,766 in 2017), which is almost a pure monopoly. With HRRs it is about 2,900 in 1999 and rises to 3,362 in 2017. The community-detection markets sit in between, at about 5,300 in 1999 and 5,517 in 2017. After 1999 all three lines are fairly flat, with a slow upward drift under the HRR and community-detection definitions.

The high values in 1998 are a data issue and not a real change. Only 1,967 hospitals report in 1998, compared with about 3,100 in 1999, and missing hospitals make every market look more concentrated than it is. I therefore read the trend from 1999 onward.

1(b) Why the definition matters

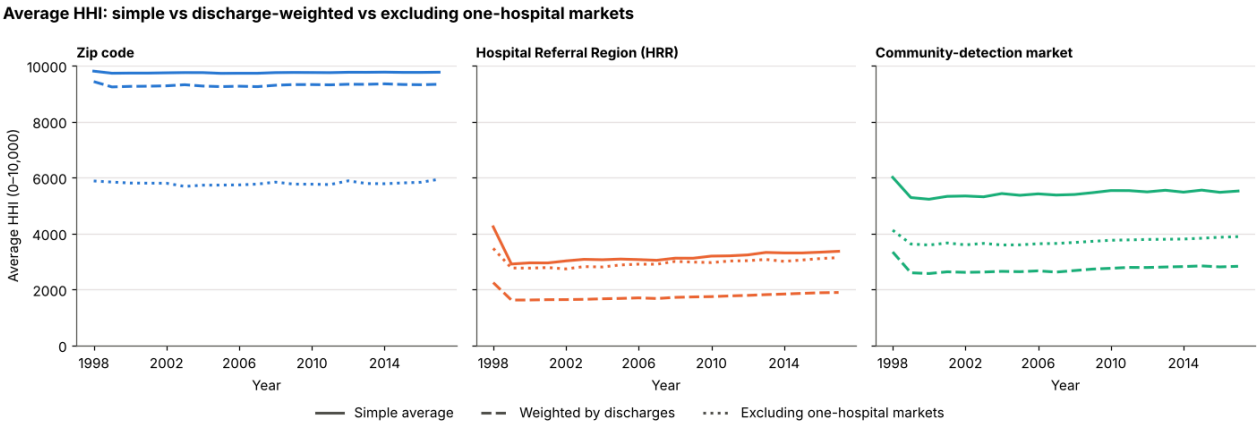
Answer. HHI is built from market shares, and a share only means something relative to the group of hospitals in the denominator. A narrow market has few hospitals in it, so each one has a large share and HHI is high. A wide market has many hospitals, so shares and HHI are low. The same hospitals can look like monopolists or like competitors depending on where the boundary is.

- **Zip code** is too narrow. In 2017, 94% of zip-code markets contain a single hospital, so HHI is 10,000 almost by construction. Patients routinely cross zip-code lines for hospital care, so this overstates market power.
- **HRR** is wide. HRRs are built around where patients are referred for major specialized surgery, and the median HRR has 5 hospitals. Two hospitals at opposite ends of an HRR may be far too distant to be substitutes for a routine delivery or an emergency, so this can understate market power.
- **Community-detection** markets are drawn from actual patient flows and land in the middle (median of 3 hospitals per market).

There is no single correct definition because the right market is the set of hospitals that patients and insurers would actually switch between, and that set is different for different services, different patients, and different insurers building a network. Any fixed geographic boundary is an approximation: it treats hospitals just outside the line as no competition at all and hospitals inside it as equal competitors. The community-detection measure is the most defensible of the three because it is based on observed behavior, but the honest conclusion is that concentration should be reported under more than one definition. Here, even the most generous definition (HRR) gives an average HHI above 2,500, a level usually treated as highly concentrated.

1(c) The problem with a simple average

year		1998	2017
Zip code	Simple average	9803	9766
	Weighted by discharges	9429	9335
	Excluding one-hospital markets	5876	5938
Hospital Referral Region (HRR)	Simple average	4238	3362
	Weighted by discharges	2241	1889
	Excluding one-hospital markets	3466	3135
Community-detection market	Simple average	6006	5517
	Weighted by discharges	3338	2825
	Excluding one-hospital markets	4116	3886




```
Zip code: 94% of markets have one hospital, but they hold 84% of discharges
Hospital Referral Region (HRR): 3% of markets have one hospital, but they hold 1% of discharges
Community-detection market: 27% of markets have one hospital, but they hold 4% of discharges
```

Answer. Both adjustments lower the average, and under the two broader definitions the drop is large. For 2017:

Definition	Simple average	Weighted by discharges	Excluding one-hospital markets
Zip code	9,766	9,335	5,938
HRR	3,362	1,889	3,135
Community-detection	5,517	2,825	3,886

The simple average counts each market once, whatever its size. Under the community-detection definition, 27% of markets in 2017 have one hospital and so an HHI of 10,000, but those markets account for only 4% of discharges. The simple average answers "how concentrated is the typical market?", while the weighted average answers "how concentrated is the market that the typical patient uses?". What the plain average leaves out is market size: most patients are treated in large markets that are much less concentrated than the many small ones.

Weighting cuts the community-detection average roughly in half (5,517 to 2,825) and moves the HRR average from "highly concentrated" to "moderately concentrated" (1,889). Excluding one-hospital markets matters most for zip codes, where nearly every market is a single hospital, and least for HRRs, where only 3% are. Neither adjustment removes the conclusion that hospital markets are concentrated: the discharge-weighted community-detection HHI is still above 2,500, and the gentle upward trend after 1999 is still there.

1(d) A sanity check on the join

```
Zip code: 52,390 market-years, share sums range 1.000000 to 1.000000  -> PASS
Hospital Referral Region (HRR): 6,064 market-years, share sums range 1.000000 to 1.000000  -> PASS
Community-detection market: 11,027 market-years, share sums range 1.000000 to 1.000000  -> PASS
```

```
Careless join, share sums within market-year: min 0.001, mean 0.055, max 1.000
Market-years passing the sum-to-1 check: 0.0%
Average HHI, correct join:  5,447
Average HHI, careless join: 26
```

Answer. The check passes for all three definitions: in every one of the 52,390 zip, 6,064 HRR, and 11,027 community-detection market-years, shares sum to exactly 1.

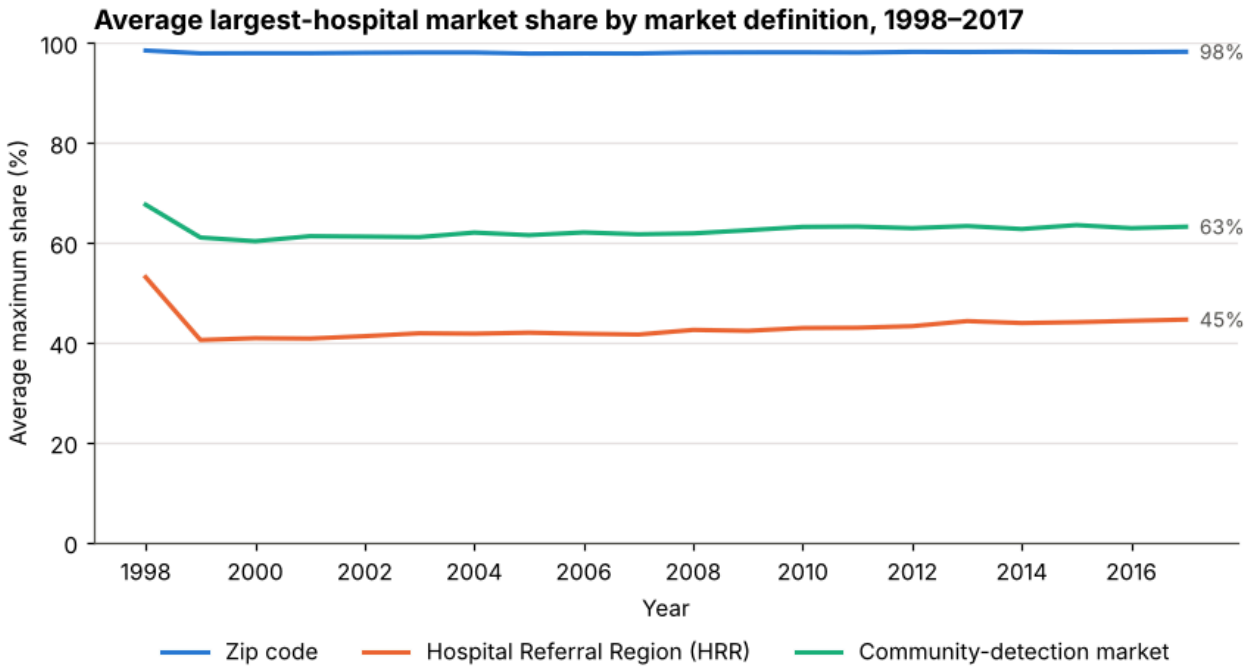
To see what the check protects against, I deliberately made a common mistake: joining market totals on the market identifier alone and leaving out the year. Each hospital-year is then divided by its market's discharges added up across all twenty years. Shares within a market-year sum to 0.055 on average instead of 1, no market-year passes the check, and the average HHI falls from 5,447 to 26. That would make hospital markets look almost perfectly competitive.

In general, a careless join breaks the link between the numerator and the denominator. If the denominator is too large (extra years or extra hospitals included), shares are too small, they sum to less than 1, and HHI is understated. Because shares are squared, the error is magnified. If rows are duplicated by the join, shares sum to more than 1 and HHI is overstated. If hospitals are matched to the wrong market's total, sums are off in both directions.

One limit of the check: it confirms that the arithmetic is consistent, not that the assignment is right. If a hospital is placed in the wrong market before the totals are computed, shares in that market still sum to 1, and the HHI is wrong anyway. So the check is necessary but does not replace looking at whether hospitals are in sensible markets.

1(e) Maximum share instead of HHI

year	Zip code	Hospital Referral Region (HRR)	Community-detection market
1998	98.4	53.1	67.6
1999	97.9	40.5	61.0
2000	97.9	40.9	60.3
2001	97.9	40.8	61.3
2002	98.0	41.3	61.2
2003	98.0	41.8	61.1
2004	98.0	41.8	62.0
2005	97.8	42.0	61.5
2006	97.8	41.8	62.0
2007	97.8	41.6	61.7
2008	98.0	42.5	61.8
2009	98.1	42.3	62.5
2010	98.1	42.9	63.1
2011	98.0	43.0	63.2
2012	98.2	43.3	62.9
2013	98.1	44.3	63.3
2014	98.2	43.9	62.7
2015	98.1	44.1	63.5
2016	98.1	44.3	62.9
2017	98.2	44.6	63.2



	HHI 1998	HHI 2017	Max share 1998 (%)	Max share 2017 (%)
Zip code	9802.9	9765.6	98.4	98.2
Hospital Referral Region (HRR)	4237.5	3361.7	53.1	44.6
Community-detection market	6005.8	5516.8	67.6	63.2

Discharge-weighted average maximum share (%):

year	Zip code	Hospital Referral Region (HRR)	Community-detection market
1998	95.5	33.8	44.0
2017	94.8	29.0	38.2

Answer. The ranking and the overall picture do not change. In 2017 the average largest-hospital share is 98% under zip codes, 63% under community-detection markets, and 45% under HRRs, and each has drifted up slightly since 1999 (from 61% and 41% for the latter two). Under every definition the average market has a dominant hospital, so I would still call these markets concentrated. Weighting by discharges lowers the figures to 38% (community-detection) and 29% (HRR) in 2017, the same pattern as in part (c).

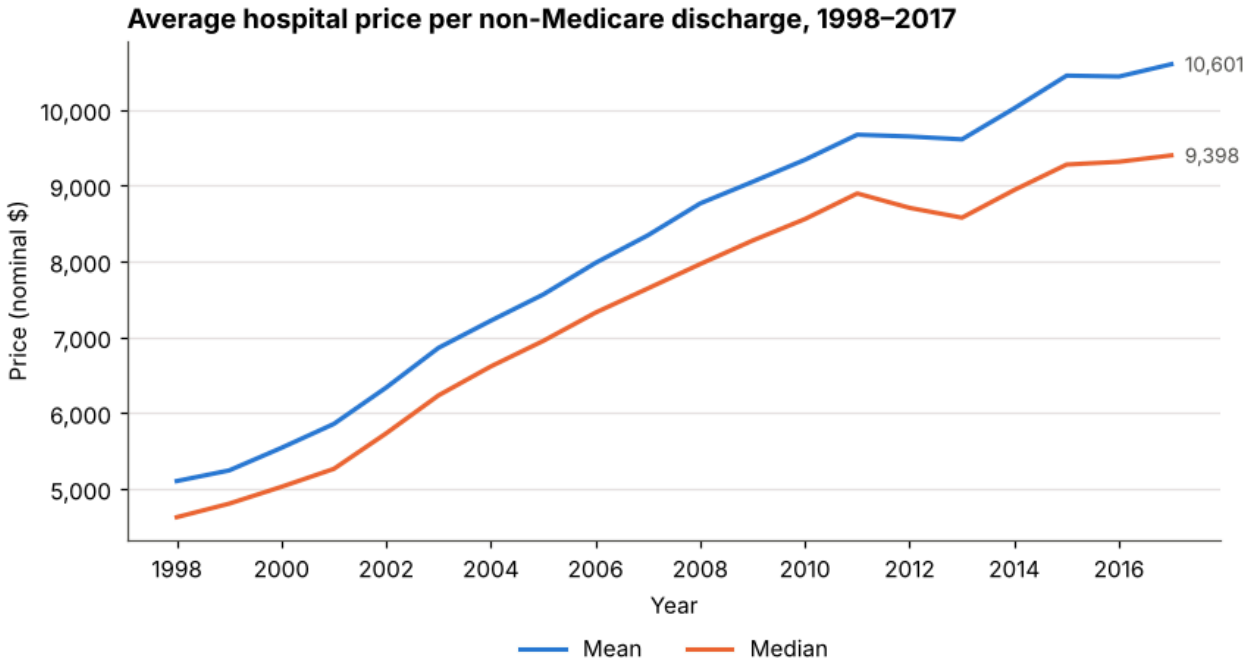
The contrast with HHI shows what each measure sees. Maximum share looks only at the leader and ignores how the rest of the market is divided. HHI uses every hospital's share. Two markets can each have a 45% leader, but if the remaining 55% belongs to one rival the HHI is 5,050, and if it is split among eleven hospitals with 5% each the HHI is 2,300. Those are very different competitive situations that maximum share cannot tell apart. In the HRR data, a 45% leader contributes about 2,000 points to HHI on its own ($0.45^2 \times 10{,}000$), so the leader explains most of the 3,362 average, but the remaining hospitals are large enough to add to it as well. The two measures agree here because concentration in these markets is mostly about one large hospital, which is also why maximum share is a reasonable quick summary but HHI is the more complete one.

Question 2 — Prices and market structure

This question uses the community-detection market (`mkt`) throughout.

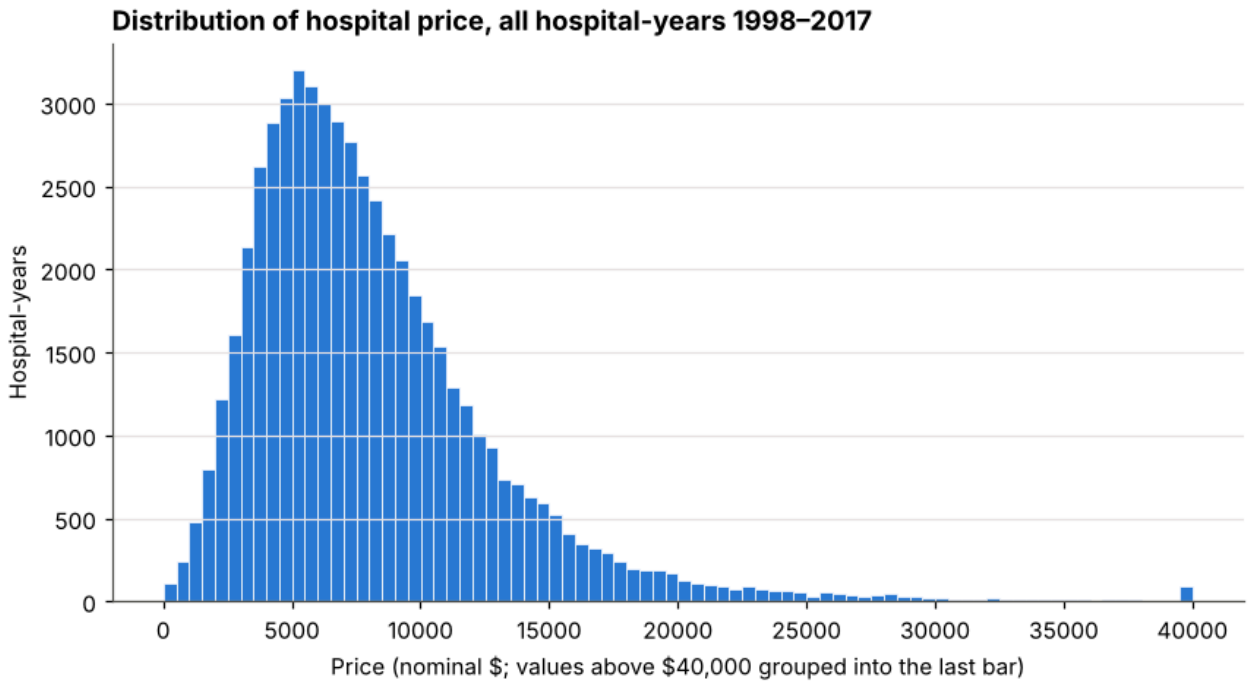
2(a) The price variable

year	Mean	Median
1998	5101	4624
1999	5241	4802
2000	5539	5025
2001	5854	5261
2002	6334	5730
2003	6858	6231
2004	7217	6613
2005	7562	6949
2006	7979	7323
2007	8343	7642
2008	8764	7964
2009	9049	8274
2010	9340	8558
2011	9671	8895
2012	9647	8705
2013	9609	8576
2014	10019	8941
2015	10450	9277
2016	10438	9314
2017	10601	9398



count	55819.0
mean	8129.0
std	5074.0
min	6.0
1%	1266.0
5%	2490.0
25%	4811.0
50%	7112.0
75%	10186.0
95%	16906.0
99%	25818.0
max	97689.0

Hospital-years with price under $1,000: 348
Hospital-years with price over $30,000: 276



year	mean	std	10%	50%	90%
1998	5101.0	3006.0	2470.0	4624.0	8013.0
2017	10601.0	6293.0	4338.0	9398.0	17659.0

Answer. The average price roughly doubles, from $5,101 in 1998 to $10,601 in 2017 (the median goes from $4,624 to $9,398). The rise is steady except for a flat stretch in 2011–2013.

The distribution is wide and skewed to the right. Across all hospital-years the median is $7,112, the middle half of hospitals lies between $4,811 and $10,186, and the standard deviation is $5,074. The range runs from $6 to $97,689. Neither end is believable as a price for a hospital stay: 348 hospital-years are below $1,000 and 276 are above $30,000. The spread also grows over time: the standard deviation doubles from $3,006 in 1998 to $6,293 in 2017, and the gap between the 10th and 90th percentile hospitals goes from about $5,500 to about $13,300.

What it captures. It is average revenue per non-Medicare discharge after discounts, so it is closer to what hospitals are actually paid than a list (chargemaster) price would be.

What it does not capture.

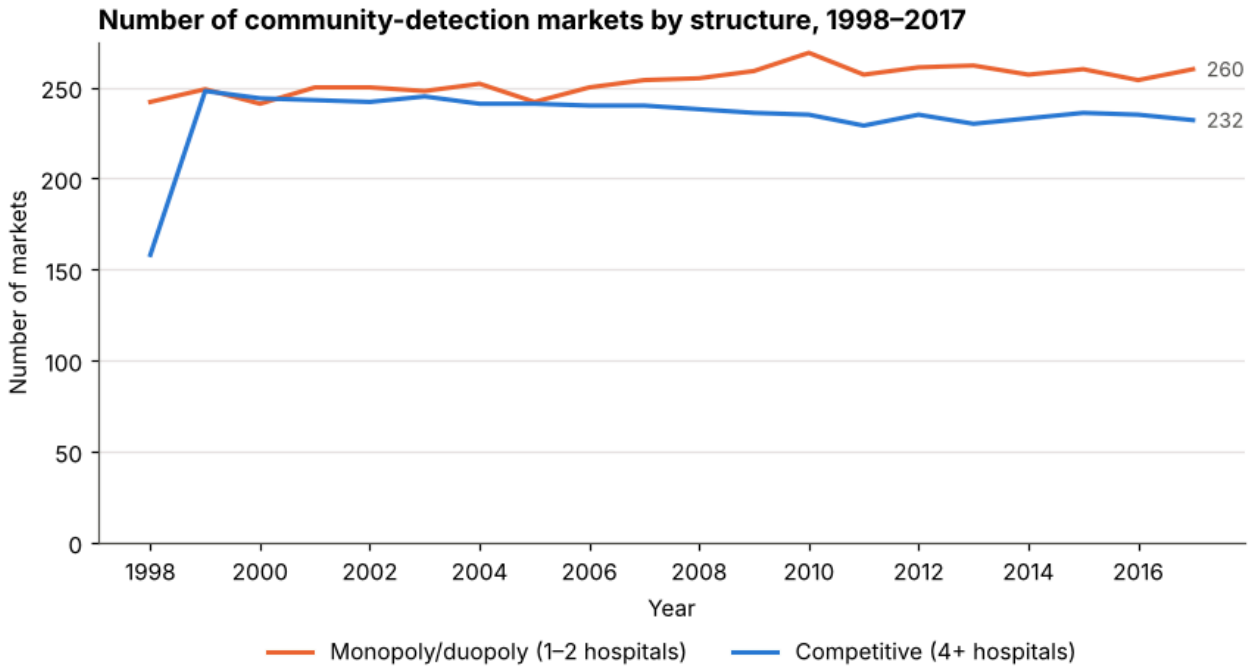
- It is not a negotiated price for any particular service. It is a ratio built from accounting totals.
- It is not adjusted for case mix. A hospital treating sicker patients or doing more complex procedures has a higher "price" even if it charges the same for the same service.
- "Non-Medicare" combines private insurance, Medicaid, and self-pay patients, so differences in payer mix move the number.

- It is built from cost-report data, which contain reporting errors. That is where the $6 and $97,689 values come from.
- It is in nominal dollars.

What that means for the trend. Consumer prices rose by roughly 50% over this period, so a doubling in nominal terms is closer to a 40% real increase. Some of the remaining increase may reflect more intensive treatment per stay and shifts in payer mix rather than higher prices for the same care. Because of the outliers and the skew, the mean sits about $500 to $1,200 above the median in every year, and the median is the safer summary. The direction of the trend is credible; the exact size is not.

2(b) Concentrated vs competitive markets

year	Competitive (4+ hospitals)	Monopoly/duopoly (1–2 hospitals)	Three hospitals (neither group)
1998	158	242	64
1999	248	249	64
2000	244	241	71
2001	243	250	70
2002	242	250	69
2003	245	248	68
2004	241	252	72
2005	241	242	63
2006	240	250	60
2007	240	254	57
2008	238	255	60
2009	236	259	61
2010	235	269	58
2011	229	257	69
2012	235	261	63
2013	230	262	64
2014	233	257	61
2015	236	260	60
2016	235	254	61
2017	232	260	59



Answer. The two groups are similar in number and move slowly in opposite directions. Monopoly or duopoly markets go from 249 in 1999 to 260 in 2017, and competitive markets

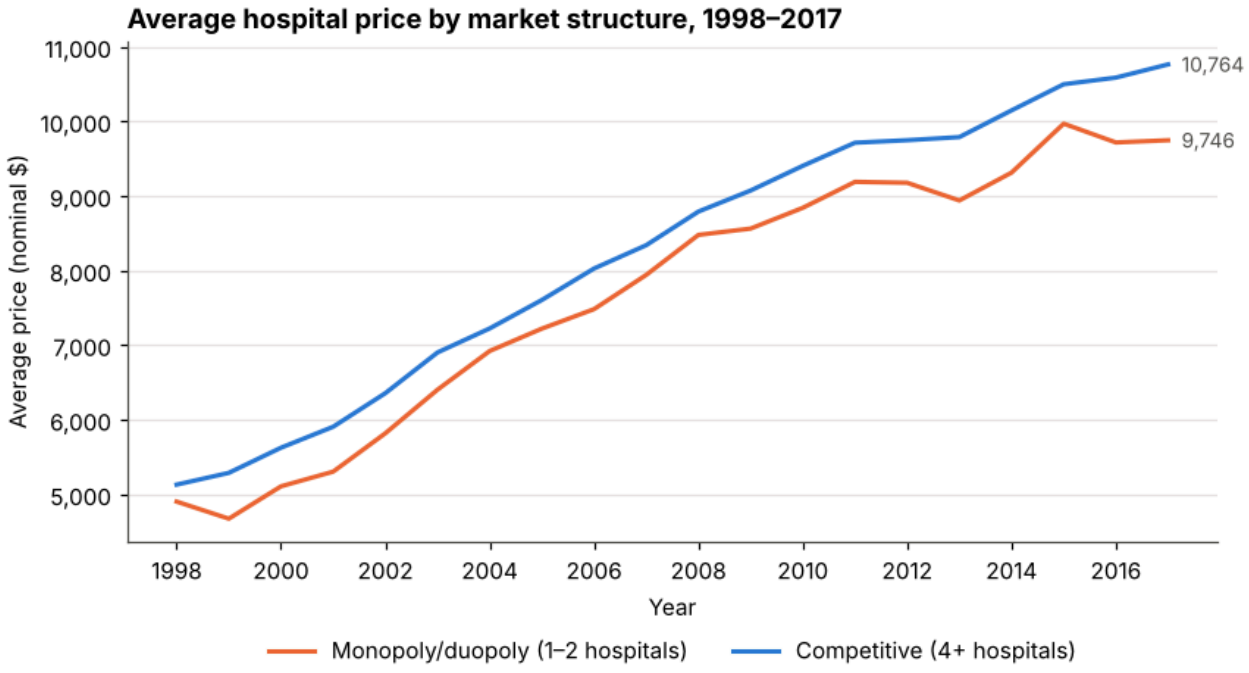
(four or more hospitals) go from 248 to 232. About 60 markets a year have exactly three hospitals and belong to neither group. In 2017, 47% of community-detection markets (260 of 551) have one or two hospitals.

The low count of competitive markets in 1998 (158) is the same reporting gap noted in Question 1: with a third of hospitals missing that year, markets that really have four or more hospitals appear to have fewer.

Two cautions. First, the shift is gradual, a net move of roughly ten to fifteen markets over eighteen years. Second, this counts hospitals, not owners. Two hospitals that merge but keep separate provider numbers still count as two competitors here, so the figure understates how much consolidation actually occurred.

2(c) Price by market structure

year	Monopoly/duopoly (1–2 hospitals)	Competitive (4+ hospitals)	Difference (concentrated − competitive)
1998	4901	5129	-228
1999	4672	5287	-615
2000	5106	5624	-518
2001	5303	5905	-602
2002	5817	6355	-538
2003	6402	6901	-499
2004	6921	7223	-303
2005	7221	7605	-384
2006	7482	8027	-545
2007	7941	8340	-398
2008	8477	8790	-313
2009	8562	9072	-510
2010	8841	9404	-563
2011	9189	9713	-525
2012	9176	9747	-571
2013	8939	9789	-850
2014	9312	10147	-835
2015	9969	10498	-529
2016	9717	10586	-869
2017	9746	10764	-1017



structure	Monopoly/duopoly (1–2 hospitals)	Competitive (4+ hospitals)
hospital_years	7116	38247
mean_price	7731	8199
median_price	6787	7189
mean_beds	151	219
mean_hospital_discharges	7475	10222
mean_market_discharges	11915	134407

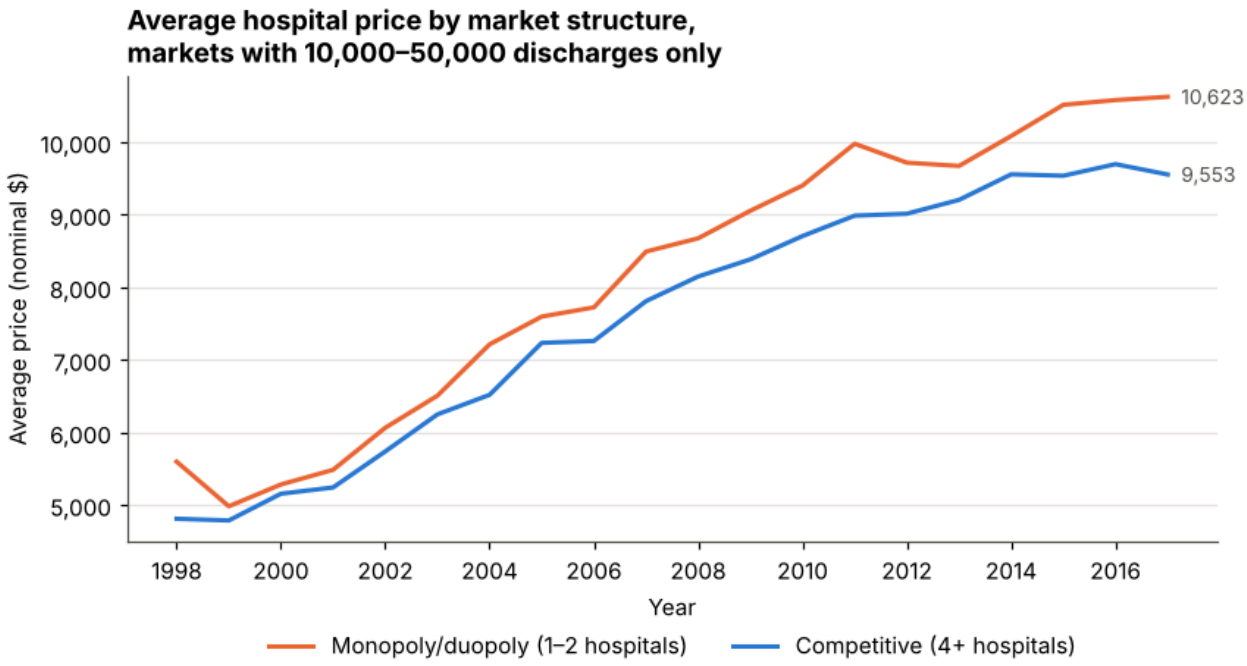
Average gap over all years: $-561

Answer. In the raw comparison, concentration is associated with **lower** prices. Hospitals in monopoly or duopoly markets have a lower average price than hospitals in competitive markets in every year. The gap averages $561 and widens over time: in 2017 it is $9,746 against $10,764, a difference of $1,017.

This is the opposite of what market power predicts, and the reason is that the two groups are very different places. Hospitals in concentrated markets average 151 beds and sit in markets with about 12,000 discharges a year. Hospitals in competitive markets average 219 beds and sit in markets with about 134,000 discharges a year. The comparison is mostly small-town hospitals against big-city hospitals, and big-city hospitals have higher wages and costs, more teaching and specialty care, and more complex cases, all of which raise this price measure. So there is a correlation, but it reflects market size at least as much as competition.

2(d) The market-size confound

structure	Monopoly/duopoly (1–2 hospitals)	Competitive (4+ hospitals)	Difference (concentrated − competitive)
year			
1998	5598	4812	787
1999	4983	4788	195
2000	5284	5156	128
2001	5486	5243	243
2002	6065	5739	326
2003	6507	6249	258
2004	7214	6519	695
2005	7596	7234	362
2006	7723	7260	463
2007	8492	7809	682
2008	8674	8148	526
2009	9054	8384	670
2010	9400	8705	695
2011	9977	8986	990
2012	9716	9013	703
2013	9671	9203	467
2014	10083	9555	527
2015	10513	9536	977
2016	10578	9696	882
2017	10623	9553	1070




```
Average price gap, concentrated minus competitive ($):
All markets                                -561.0
Markets with 10,000–50,000 discharges     582.0

By period:
           All markets  Markets with 10,000–50,000 discharges
year
1998–2007       -463.0                                  414.0
2008–2017       -658.0                                  751.0

Market-years and average size in each comparison:
                                                        hospital_years  markets  mean_market_discharges  mean_beds
                          structure
10,000–50,000 discharges Competitive (4+ hospitals)               9720      223                   32979        142
                         Monopoly/duopoly (1–2 hospitals)          3515      236                   17843        200
All markets              Competitive (4+ hospitals)              38247      304                  134407        219
                         Monopoly/duopoly (1–2 hospitals)          7116      416                   11915        151
```

Answer. Limiting the comparison to markets with 10,000 to 50,000 discharges a year reverses the sign. Hospitals in monopoly or duopoly markets are now **more** expensive in every year, by $582 on average, compared with $561 cheaper in the full sample. The gap grows over time, from $414 in 1998–2007 to $751 in 2008–2017, and reaches $1,070 in 2017 ($10,623 against $9,553).

Market size is a confound because it affects both things being compared. Small markets can only support one or two hospitals because of fixed costs, so small size causes concentration. Small, rural markets also have lower prices for reasons unrelated to competition: lower wages and costs, less complex cases, and less specialty care. In the raw data the "concentrated" group is mostly small markets and the "competitive" group is mostly large ones, so the low-price effect of being small is credited to concentration and hides the market-power effect. Comparing markets of similar size removes most of that, and the positive relationship predicted by theory appears.

The restriction does not make the groups identical. Within the band, concentrated markets are still smaller on average (about 17,800 discharges against 33,000), and their hospitals are

larger (200 beds against 142), since one or two hospitals are serving a whole mid-sized market. Larger hospitals tend to offer more complex services, which could push their measured price up. So the restricted comparison is better than the raw one but is still a correlation.

2(e) Is it causal?

Answer. No. The size-restricted comparison in part (d) is consistent with concentration raising prices, but it is not causal evidence.

Why not.

- *Remaining confounders.* Concentrated and competitive markets still differ in case mix, payer mix, local wages and costs, hospital size, and quality. Any of these could produce a price difference without market power playing a role.
- *Market structure is not random.* The number of hospitals in a market is an outcome of demand, costs, and past entry, exit, and merger decisions. Causation can also run the other way: high prices and margins attract entry, and a hospital that is better or lower-cost gains share and ends up in a concentrated market.
- *The price measure is noisy.* As discussed in part (a), it is not adjusted for case mix and it mixes payers.
- *The answer already changed sign once.* Going from a $561 discount to a $582 premium by controlling for one variable shows how sensitive the comparison is to what is held fixed.

What a causal claim would need. A change in concentration that is unrelated to the other things that drive price. The standard approach is to study mergers or closures: follow prices at the affected hospitals before and after the event, and compare them with similar hospitals in markets where nothing changed (a difference-in-differences design), keeping each hospital as its own comparison over time. That works best with mergers that happened for reasons unrelated to local market conditions, for example when two national chains combine. It would also need a better price measure, ideally negotiated payments from insurance claims for a fixed set of services, adjusted for case mix.

Conclusion. Once markets of similar size are compared, hospitals in concentrated markets charge about 5% to 11% more in 2013–2017. That matches what theory predicts and what merger studies generally find, so I read it as suggestive evidence of market power, but this data alone cannot establish that concentration is the cause.